

Lecture 2: Data-Centric Methods

Dr. LI Yinqiao

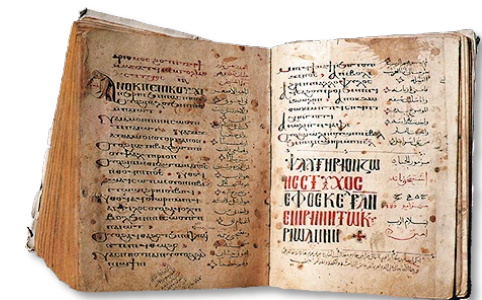
li.yin.qiao.2012@hotmail.com

<https://liyinqiao.com>

Presenter Video

*Available in English,
Cantonese and Mandarin*

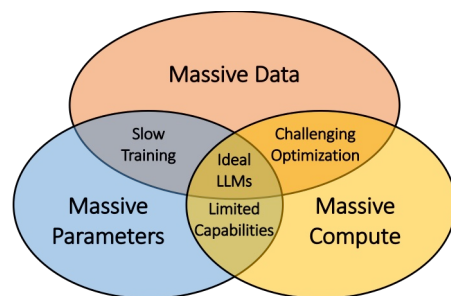
>> *“In the beginning was the Word, and the Word was with God, and the Word was God.”*
- *The Gospel According to John, 1:1*



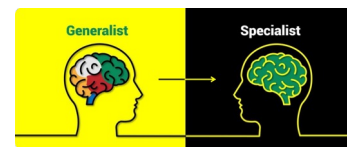
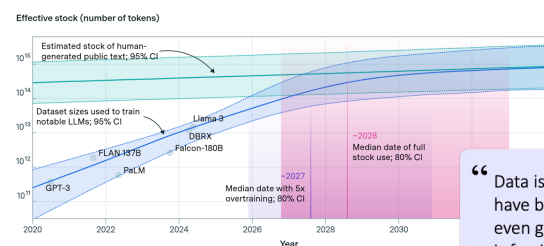
NEW TESTAMENT
(Part of the Bible)

Recap: Previous Lecture

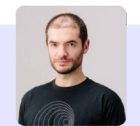
- Understand the basic definition of large language models (LLMs)
- Learn the general process of developing and deploying LLMs
- Recognize the low-resource challenges of applying LLMs to real-world tasks
- Get a bird's-eye view of low-resource task adaptation



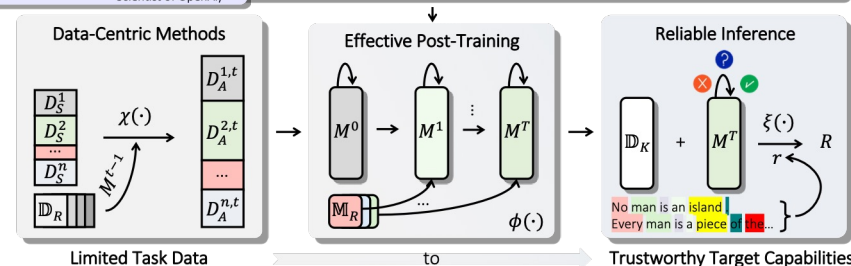
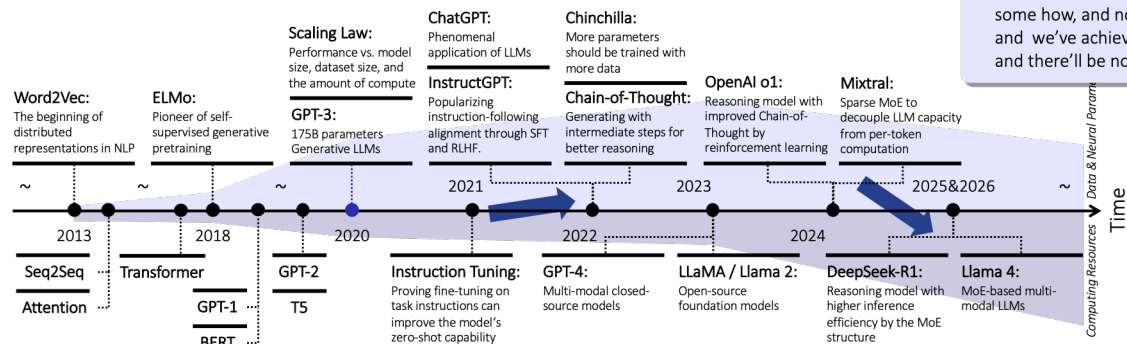
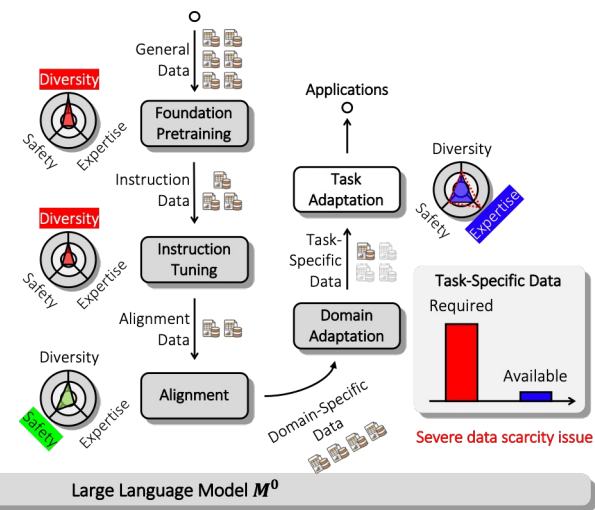
Projections of the stock of public text and data usage



“Data is not growing, because we have but one Internet. You could even go as far as to say that data is fossil fuel of AI. It was created some how, and now we use it, and we’ve achieved peak data and there’ll be no more^[1].”



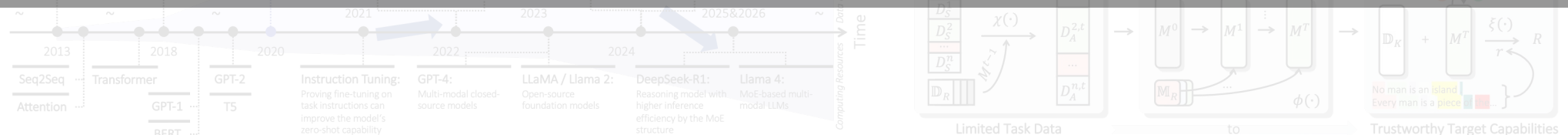
Ilya Efimovich Sutskever
(Co-founder and Chief Scientist of OpenAI)



Recap: Previous Lecture

- Understand the basic definition of large language models (LLMs)
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- Get a bird's-eye view of low-resource task adaptation

- 1 **How** can we obtain and leverage high-quality supervision?
- 2 How can we perform effective model updating under limited labeled data?
- 3 How can we improve reliability at inference time for low-resource LLMs?



CONTENT

- What Will You Learn from This Tutorial?
 - Understand the main angles to address data scarcity from a data-centric perspective ★☆☆☆^[1]
 - Learn how data augmentation, data quality evaluation and selection, and data mixing methods can support effective low-resource task adaptation ★★★

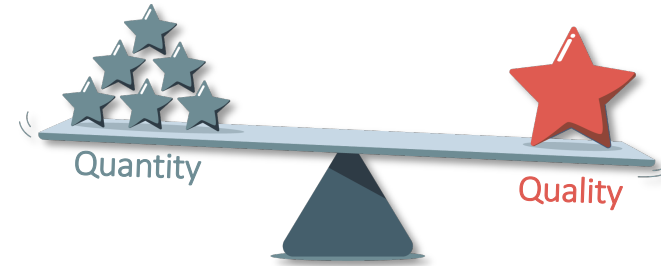
OUTLINE

- How to Get Reliable Supervision with Limited Data?
- Data Augmentation
 - Synthetic Supervision Construction
 - Semi-Supervised Expansion
- Data Quality Evaluation and Selection
 - Data-Aware Evaluation
 - Model-Dependent Evaluation
- Data Mixing
 - Static Mixing
 - Adaptive Mixing
- Summary
- Exercises
- Bibliography for Further Study
- Index
- Acknowledgements

[1] The stars indicate the difficulty level of each learning objective.

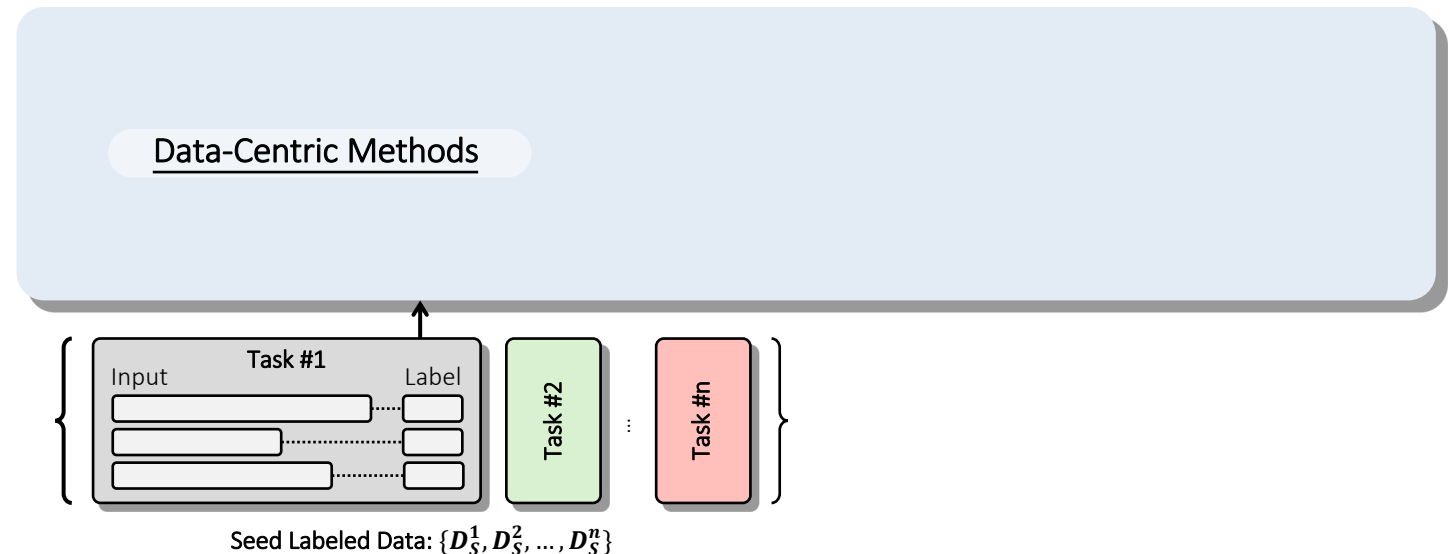
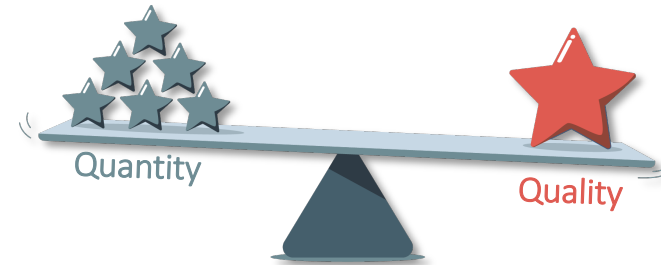
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- Data scarcity is a fundamental challenge in low-resource task adaptation.
- What kinds of supervisions do we need?
 - The more, the better!
 - Quality is everything!



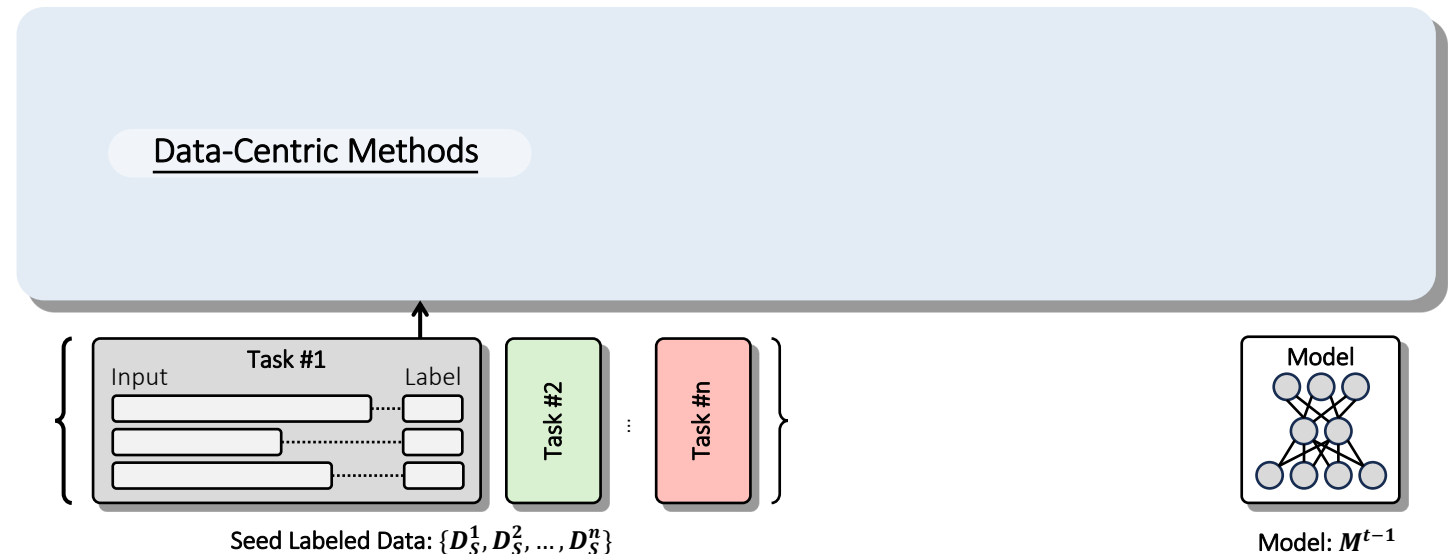
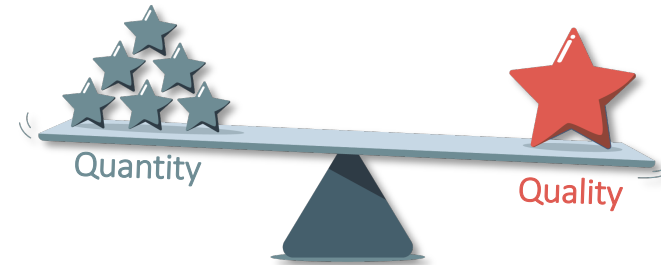
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 - For the datasets applied at the t_{th} post-training step, we can explore reliable supervisions based on:
 - Seed labeled datasets from multiple tasks:
 $\{D_S^1, \dots, D_S^n\}$



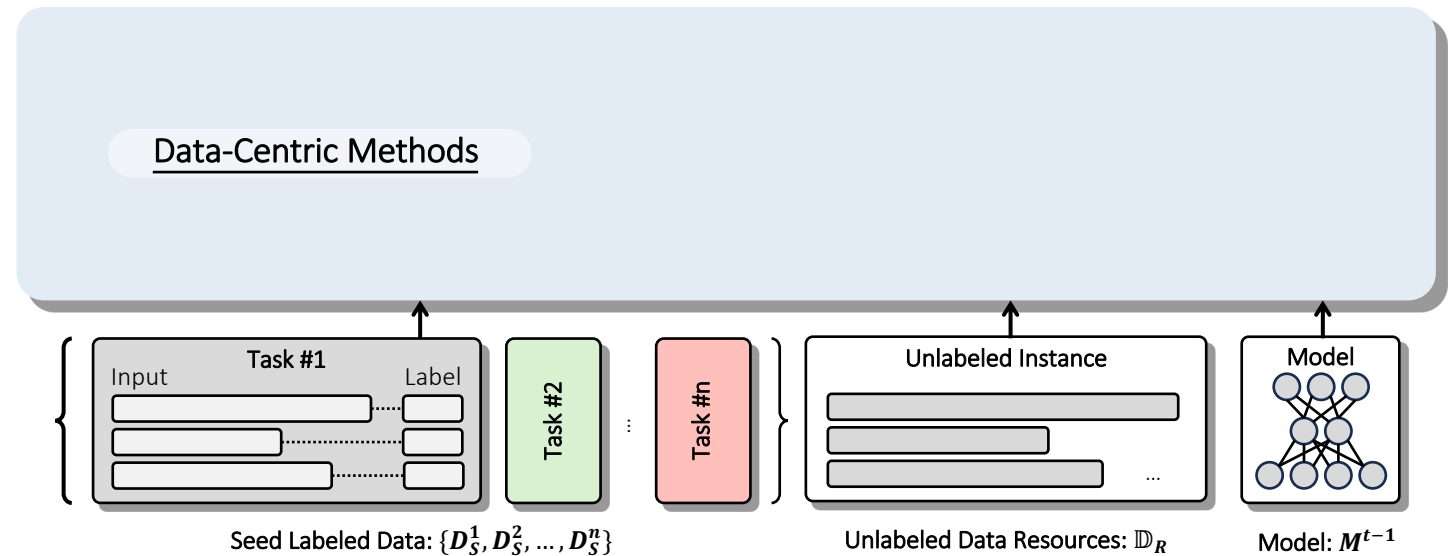
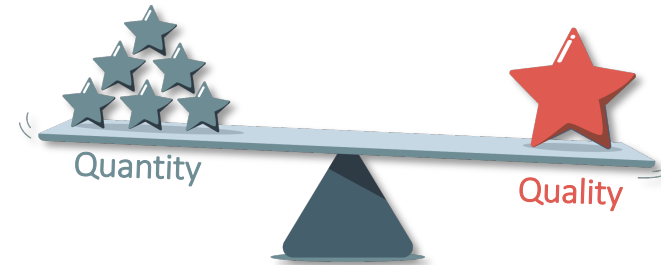
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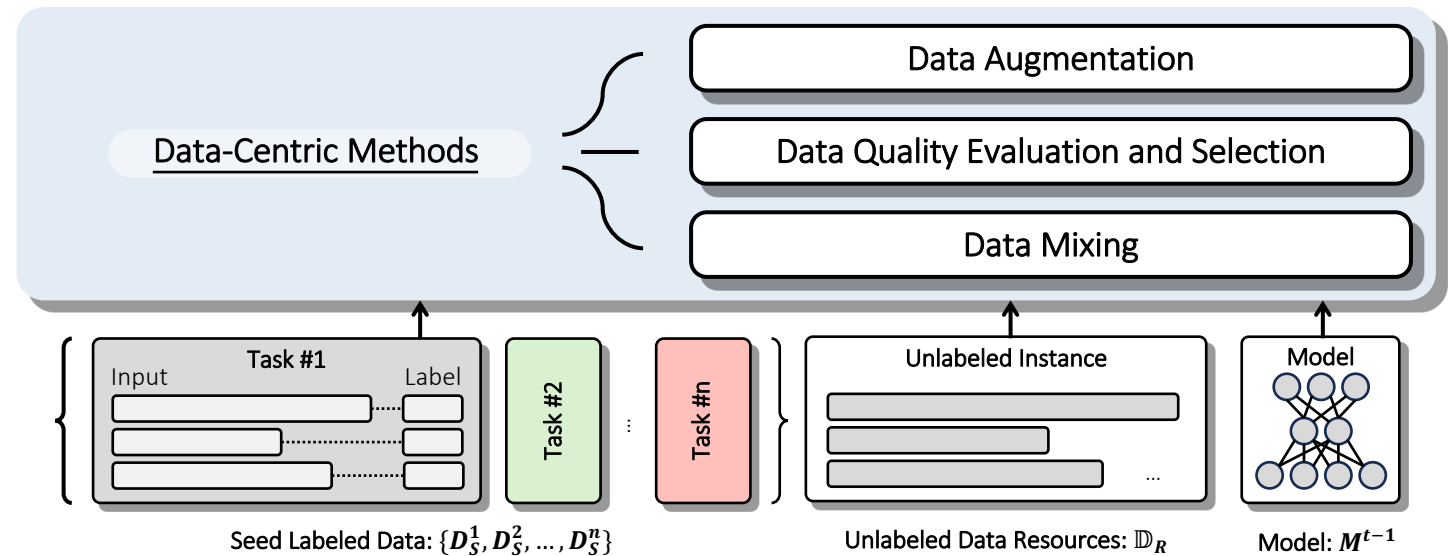
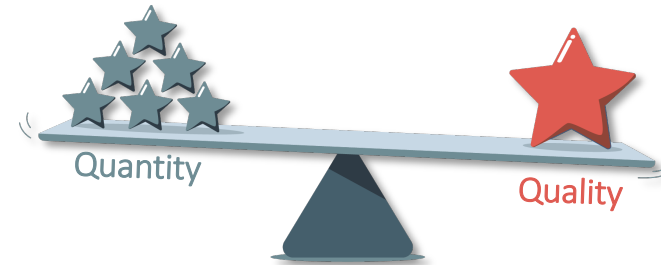
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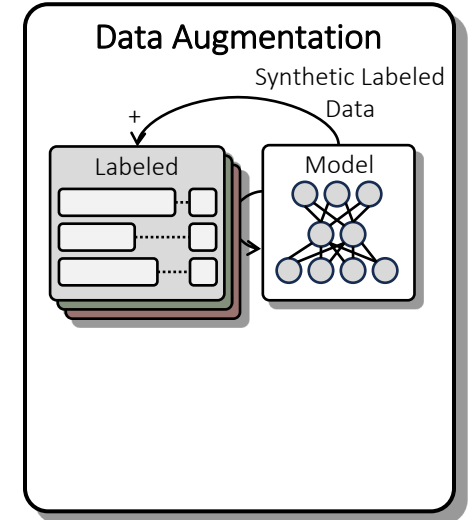
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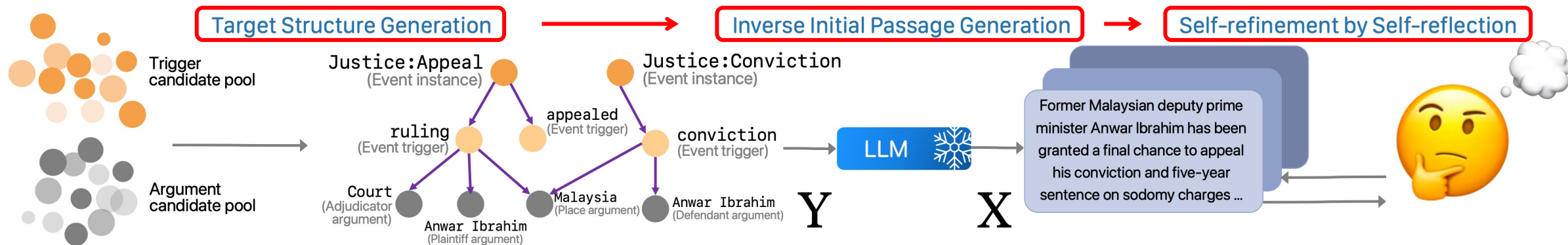
Data Augmentation

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- Synthetic supervision construction: Directly generate with generative models
 - How to **generate** data while ensuring compatibility **with target task requirements**?



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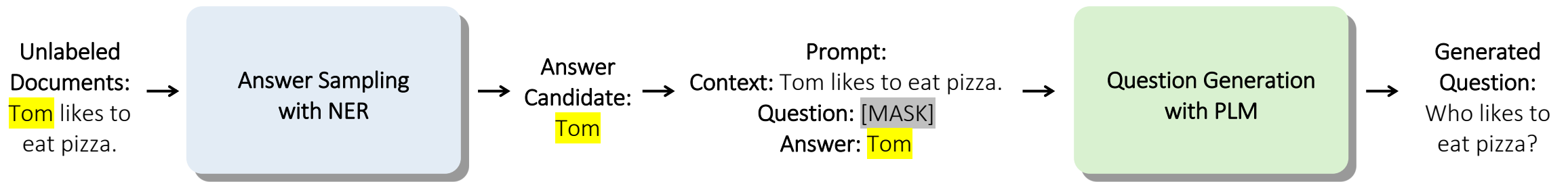
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[1] Mingyu Derek Ma, Xiaoxuan Wang, Po-Nien Kung, P. Jeffrey Brantingham, Nanyun Peng, Wei Wang: STAR: Boosting Low-Resource Information Extraction by Structure-to-Text Data Generation with Large Language Models. AAAI 2024: 18751-18759

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 - Question Answering: Sample answer first, then generate questions conditioned on the context and answers ^[2]

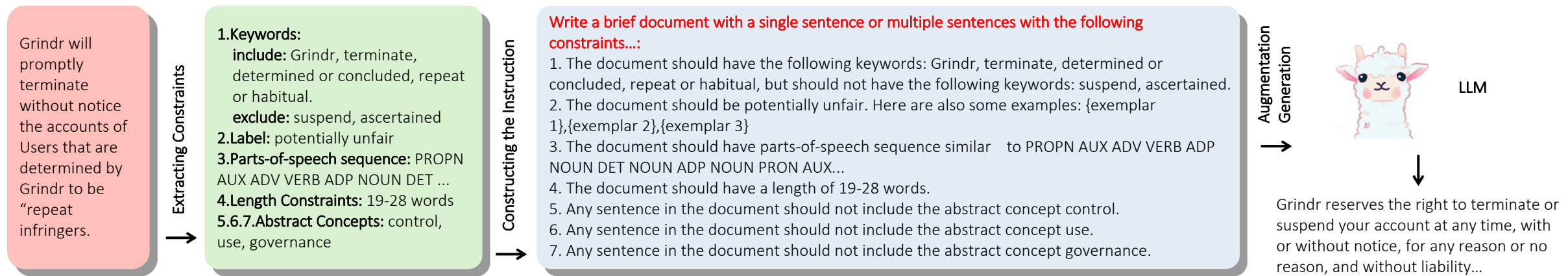


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[2] Maximilian Schmidt, Andrea Bartezzaghi, Ngoc Thang Vu: Prompting-based Synthetic Data Generation for Few-Shot Question Answering. LREC/COLING 2024: 13168-13178

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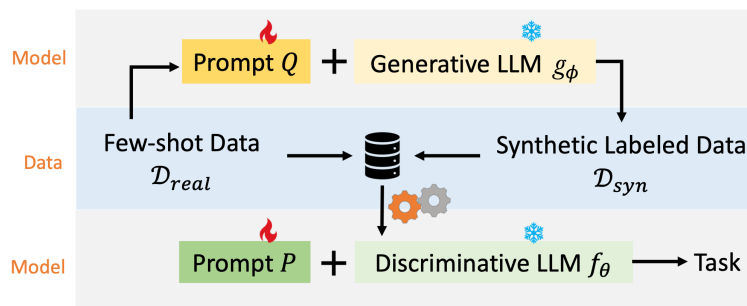
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 - Prompting: Extract constraints from instances for controllable generation ^[1]



[1] Chandra Kiran Reddy Evuru, Sreyan Ghosh, Sonal Kumar, Ramaneswaran S., Utkarsh Tyagi, Dinesh Manocha: CoDa: Constrained Generation based Data Augmentation for Low-Resource NLP. NAACL-HLT (Findings) 2024: 3754-3769

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 - Alignment: Align the data generator with the target few-shot distribution ^[2]



Step 1: training a soft prompt Q for a frozen LLM g_ϕ on the few-shot training set $\mathcal{D}_{real}$. **Step 2:** apply g_ϕ and Q to generate a synthetic labeled training set $\mathcal{D}_{syn}$. **Step 3:** With $\mathcal{D}_{real}$ and $\mathcal{D}_{syn}$, train a soft prompt P for a discriminative LLM f_θ to perform downstream tasks, while tackling issues like gradient conflicts.

$$\mathcal{L}_{\text{DawGen}}(Q) = \mathbb{E}_{l \in [1, L]} \mathcal{L}_{\text{wGen}}(Q_l) + \mathcal{L}_{\text{dist}}(Q)$$

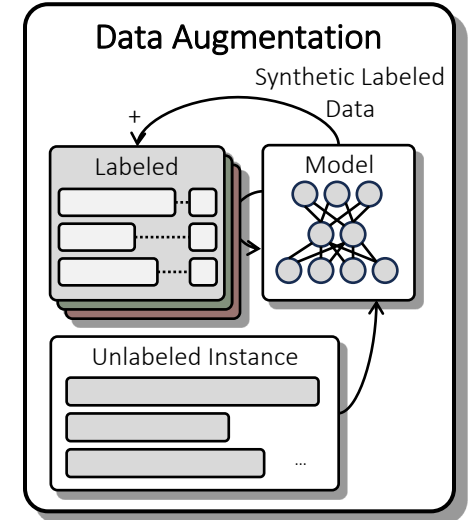
“ We propose to regularize the generator tuning objective by adding a sentence-level distribution constraint to encourage the generated sentence to align with the in-distribution data. ”

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[2] Xu Guo, Zilin Du, Boyang Li, Chunyan Miao: Generating Synthetic Datasets for Few-shot Prompt Tuning. CoLM 2024

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 - Applying unlabeled data to solve the low-resource issue is classic in NLP, such as back-translation in early NMT ^[1] and later pretrained language models.

“ Back-translation is an essential method to integrate the target side monolingual synthetic knowledge when building a state-of-the-art NMT system. Especially for low-resource language tasks, it's indispensable to augment the training data by mixing the pseudo corpus with the parallel part, in that the target side lexicon coverage is insufficient ^[2]. ”

Kazakh→English		
Ave.	Ave. z	System
72.2	0.270	online-B
70.1	0.218	NEU
69.7	0.189	rug-morfessor
68.1	0.133	online-G
67.1	0.113	talp-upc-2019
67.0	0.092	NRC-CNRC
65.8	0.066	Frank-s-MT
65.6	0.064	NICT
64.5	0.003	CUNI-T2T-transfer
48.9	-0.477	UMD
32.1	-1.058	DBMS-KU

ACL 2019
FOURTH CONFERENCE ON
MACHINE TRANSLATION (WMT19)

August 1-2, 2019
Florence, Italy

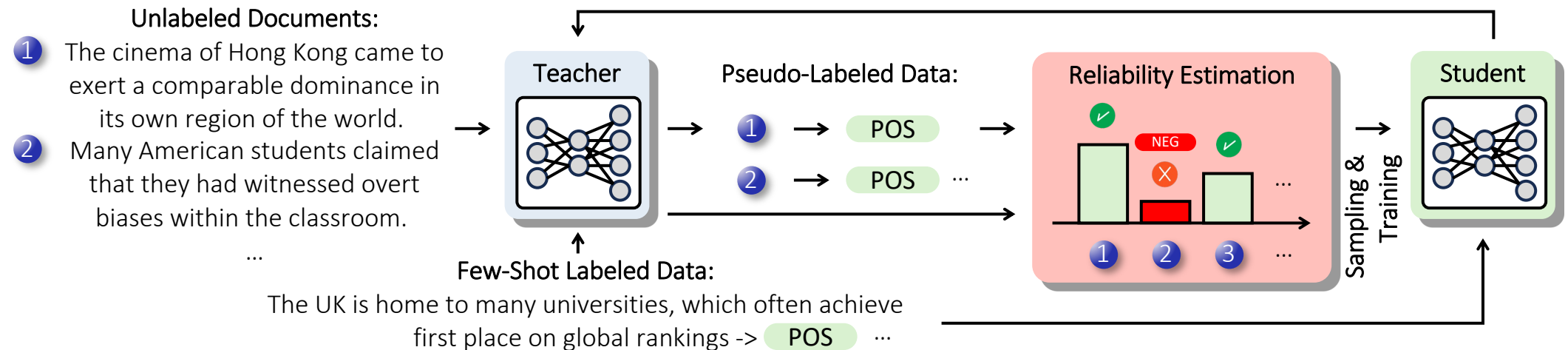
KK-EN		
System	19dev	19test
Big	10.1	11.5
+Pivot	23.4	19.8
+Sampling	26.2	28.8

[1] Rico Sennrich, Barry Haddow, Alexandra Birch: Improving Neural Machine Translation Models with Monolingual Data. ACL (1) 2016

[2] Bei Li, Yinqiao Li, Chen Xu, Ye Lin, Jiqiang Liu, Hui Liu, Ziyang Wang, Yuhao Zhang, Nuo Xu, Zeyang Wang, Kai Feng, Hexuan Chen, Tengbo Liu, Yanyang Li, Qiang Wang, Tong Xiao, Jingbo Zhu: The NiuTrans Machine Translation Systems for WMT19. WMT (2) 2019: 257-266

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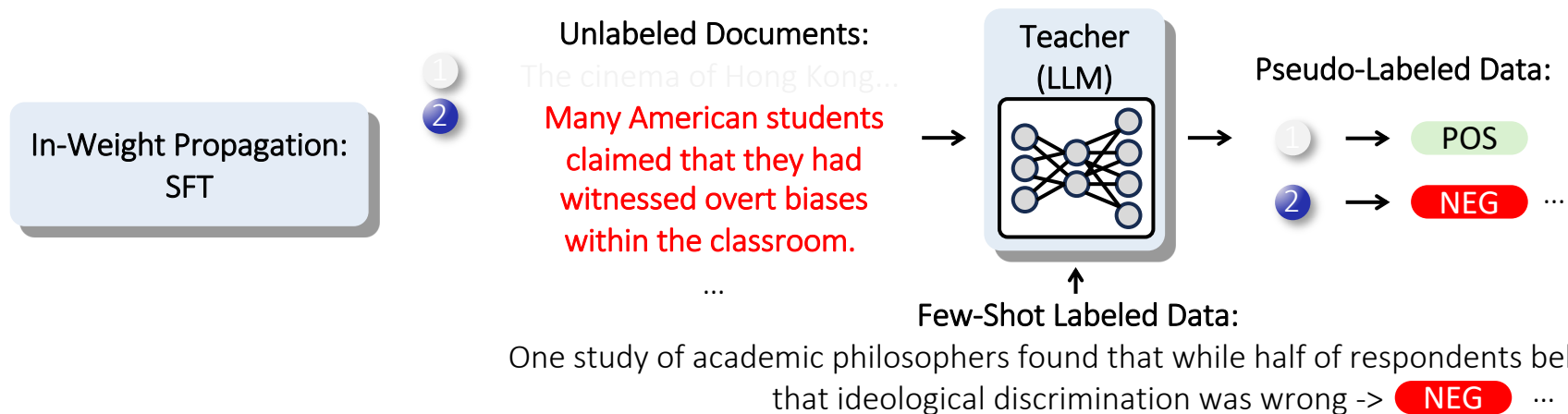


[1] Yaqing Wang, Subhabrata Mukherjee, Xiaodong Liu, Jing Gao, etc.: LiST: Lite Prompted Self-training Makes Parameter-efficient Few-shot Learners. NAACL-HLT (Findings) 2022: 2262-2281

[2] Jianing Wang, Qiushi Sun, Nuo Chen, etc.: Uncertainty-aware Parameter-Efficient Self-training for Semi-supervised Language Understanding. EMNLP (Findings) 2023: 7873-7884

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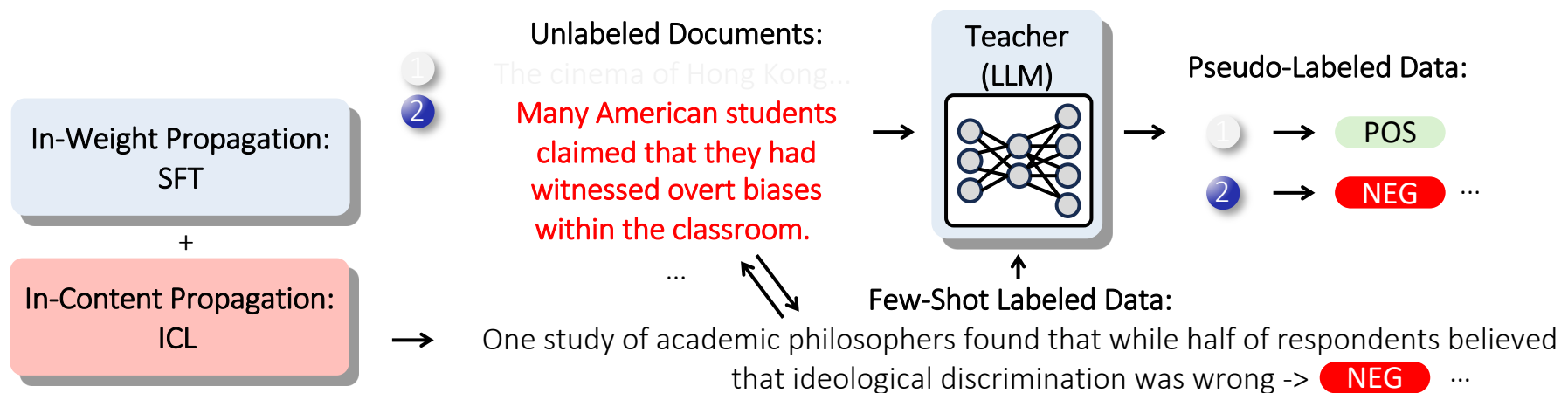
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 - Propagate-and-select framework: Transfer with both in-weight and **in-context propagation** [1]



[1] Junyu Luo, Xiao Luo, Xiusi Chen, Zhiping Xiao, Wei Ju, Ming Zhang: Semi-supervised Fine-tuning for Large Language Models. NAACL (Findings) 2025: 2795-2808

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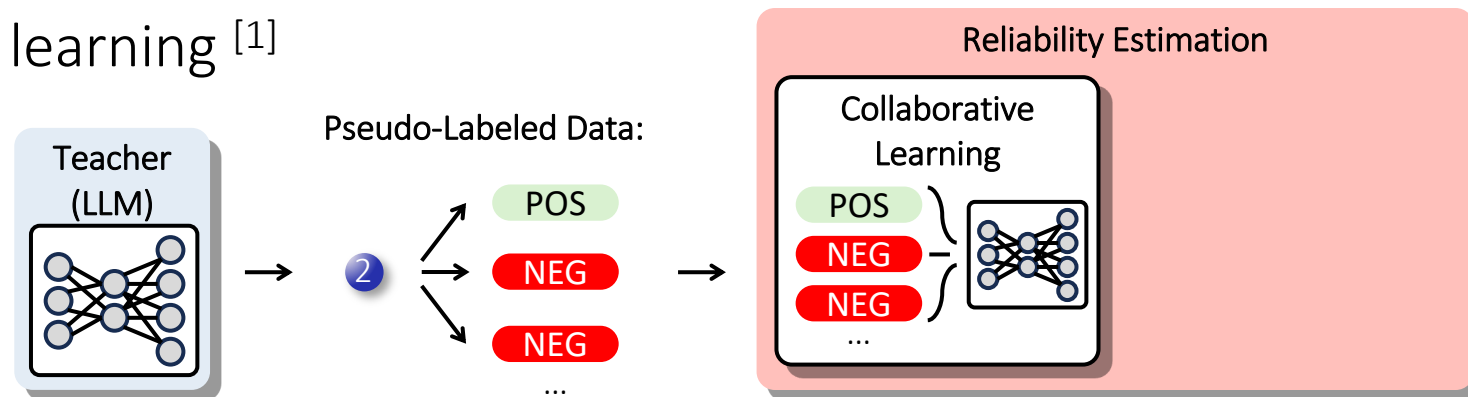
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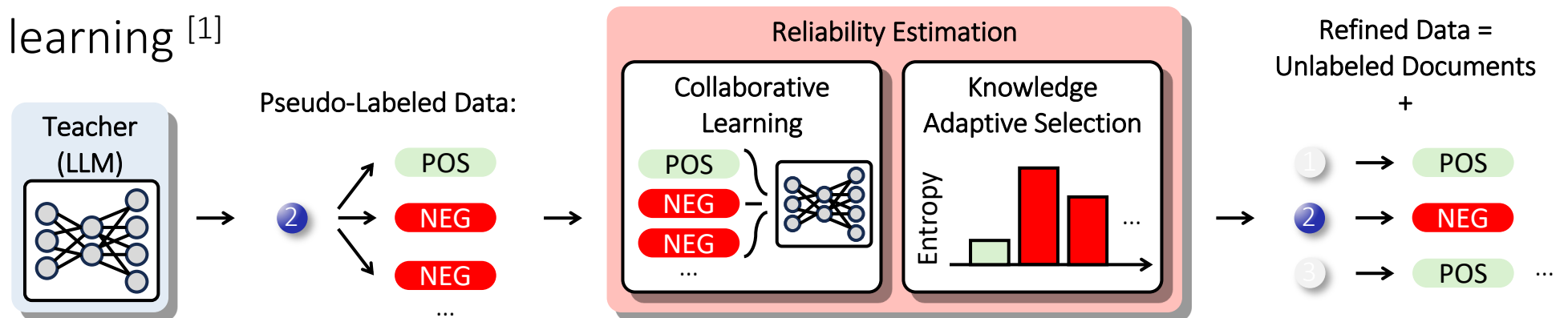
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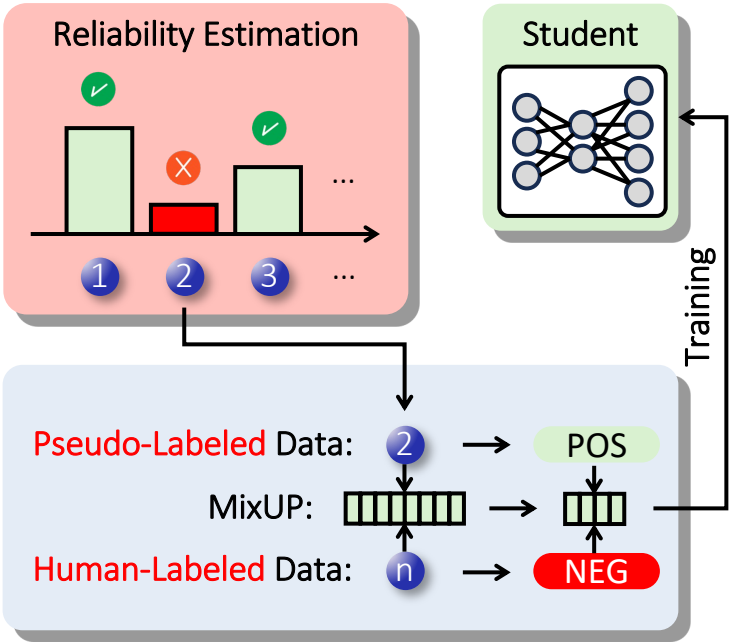


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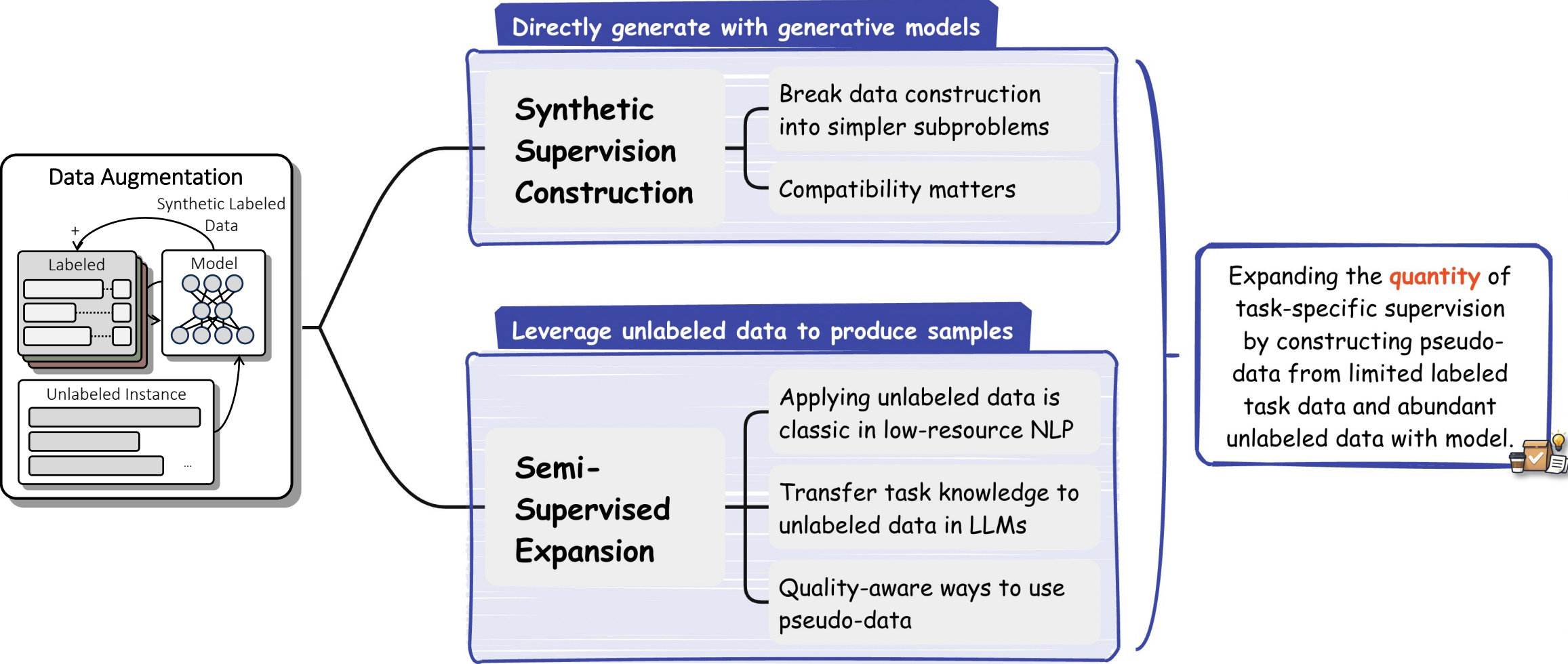
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 - Verify, don't discard: Identify unreliable pseudo-data and denoise it via confidence-aware MixUp with human-labeled data ^[1].

	RTE	SICK	SNLI-2.5k	MNLI-2.5k _m	MNLI-2.5k _{mm}
VerifyMatch (Ours)	71.03 _{2.1}	86.96 _{0.8}	82.06 _{0.3}	74.20 _{0.5}	74.10 _{0.3}
w/o mismatched PG data	68.55 _{1.5}	85.37 _{0.8}	80.85 _{0.6}	71.87 _{0.4}	72.12 _{0.8}
w/o unconfident PG data	68.34 _{1.3}	85.88 _{0.5}	79.21 _{0.4}	71.71 _{0.6}	72.13 _{0.2}
w/ Lower Weights	68.49 _{2.1}	85.59 _{1.2}	78.92 _{0.8}	72.12 _{0.5}	72.36 _{0.4}



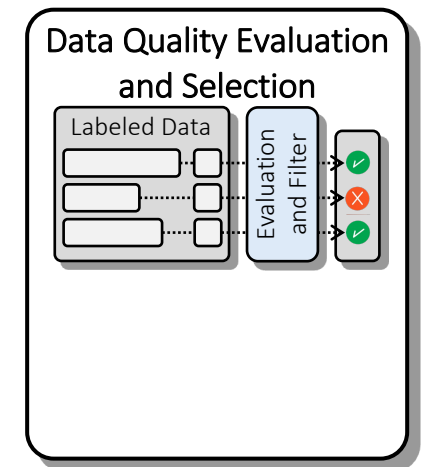
[1] Seoyeon Park, Cornelia Caragea: VerifyMatch: A Semi-Supervised Learning Paradigm for Natural Language Inference with Confidence-Aware MixUp. EMNLP 2024: 19319-19335

Takeaways



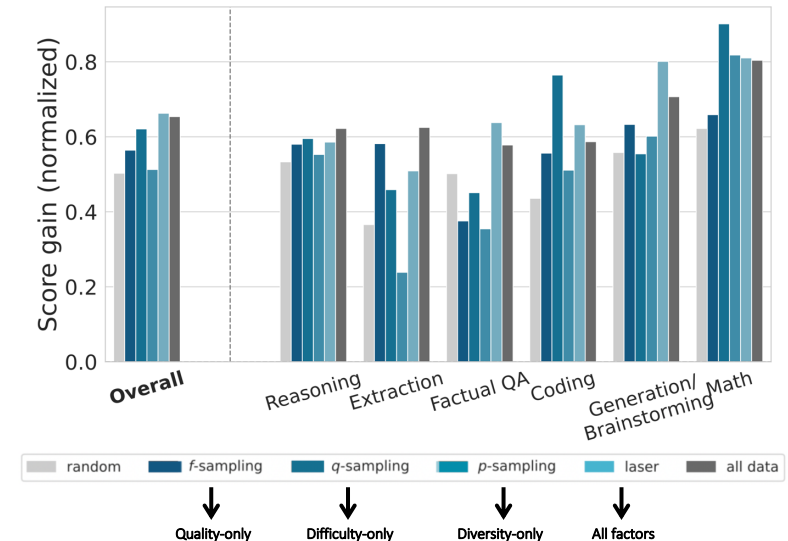
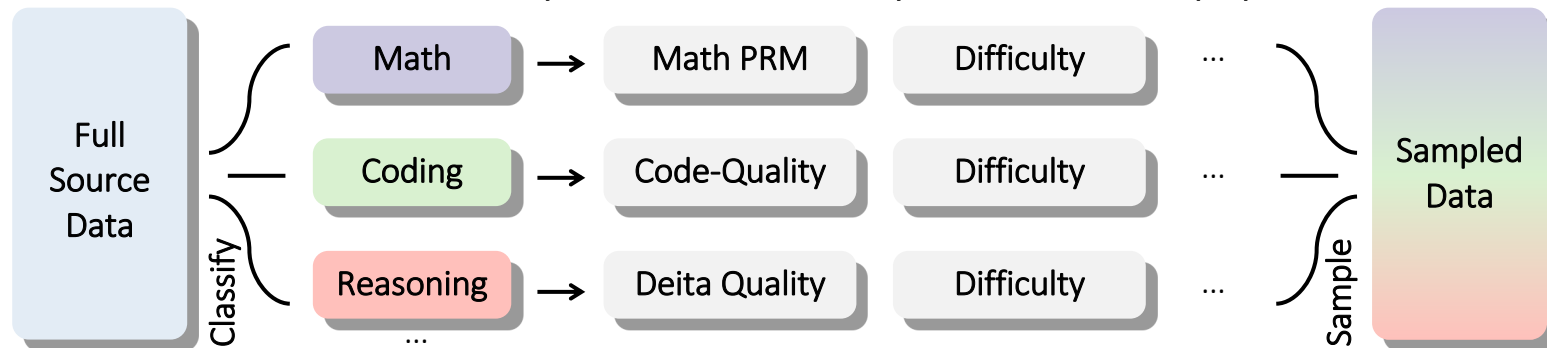
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- The quality of both real and synthetic task-specific data matters.
- Data-aware evaluation: Estimate instances using sample-level signals
 - How to **identify and handle** unreliable samples **based on the textual features**?



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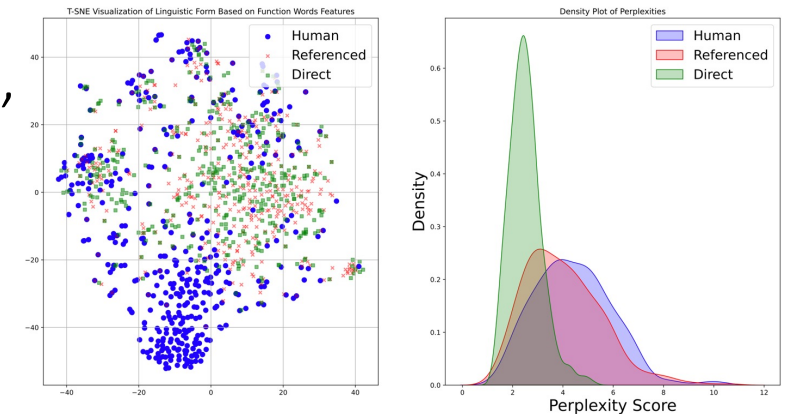
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 - Quality (correctness, instruction-following, etc.), difficulty, and diversity: A unified pipeline ^[1].



[1] Paramita Mirza, Lucas Weber, Fabian K  ch: Stratified Selective Sampling for Instruction Tuning with Dedicated Scoring Strategy. EMNLP (Findings) 2025: 19949-19974

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[2] Zhuang Li, Yuncheng Hua, Thuy-Trang Vu, Haolan Zhan, Lizhen Qu, Gholamreza Haffari: SCAR: Data Selection via Style Consistency-Aware Response Ranking for Efficient Instruction-Tuning of Large Language Models. ACL (1) 2025: 12756-12790

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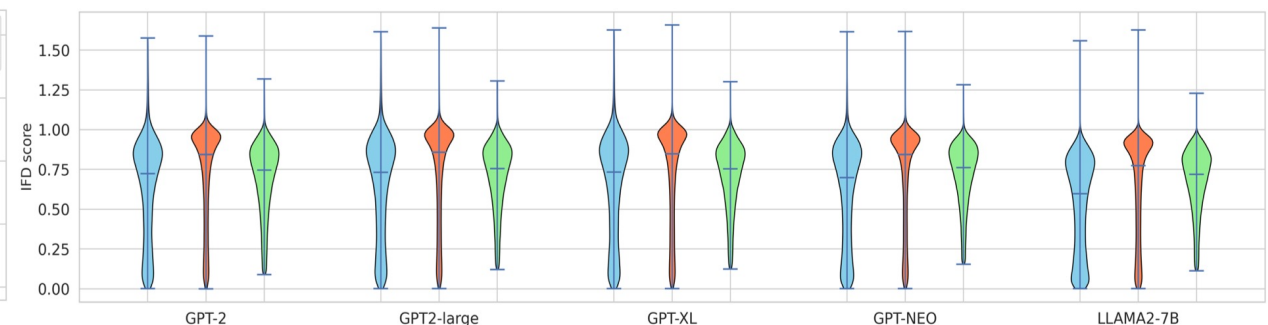
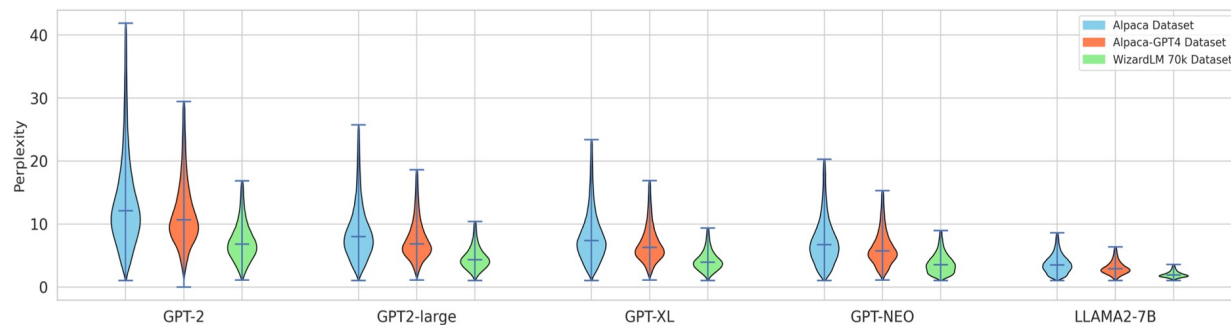
Data Curation Methods	STACKEXCHANGE		
	Stylometric Analysis	Data Quality	CODELLAMA-7B Performance
	Std. TTR ↓ / Std. PPL($y x$) ↓	Helpfulness / Correctness	Avg. Pass@1 / Avg. Pass@10
Human Response	24.23 / 0.33	3.29 / 3.70	26.56 / 41.63
GPT-3.5-TURBO			
Referenced	8.16 / 0.33	3.44 / 3.70	29.82 / 46.89
Direct	8.14 / 0.30	3.32 / 3.45	31.00 / 47.12
LLAMA2-70B-CHAT			
Referenced	11.90 / 0.36	3.14 / 3.54	29.82 / 44.03
Direct	13.52 / 0.28	3.18 / 2.71	30.89 / 45.31
LLAMA2-13B-CHAT			
Referenced	7.46 / 0.27	2.65 / 2.68	26.61 / 41.91
Direct	8.86 / 0.28	1.85 / 1.70	26.42 / 40.00

[1] Paramita Mirza, Lucas Weber, Fabian Küch: Stratified Selective Sampling for Instruction Tuning with Dedicated Scoring Strategy. EMNLP (Findings) 2025: 19949-19974

[2] Zhuang Li, Yuncheng Hua, Thuy-Trang Vu, Haolan Zhan, Lizhen Qu, Gholamreza Haffari: SCAR: Data Selection via Style Consistency-Aware Response Ranking for Efficient Instruction-Tuning of Large Language Models. ACL (1) 2025: 12756-12790

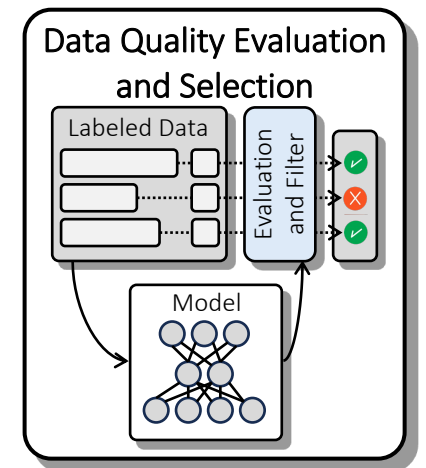
Data Quality Evaluation and Selection

- Data quality is a vital factor for LLM adaptation, especially for the tasks with scarce data.
- The quality of both real and synthetic task-specific data matters.
- Data-aware evaluation: Estimate instances using sample-level signals
 - How to **identify and handle** unreliable samples **based on the textual features**?
 - Evaluation **efficiency** is crucial given the potentially large pool of candidates.
 - Lightweight evaluation: Lightweight proxy models can effectively replace larger models in rapid data filtering ^[1]



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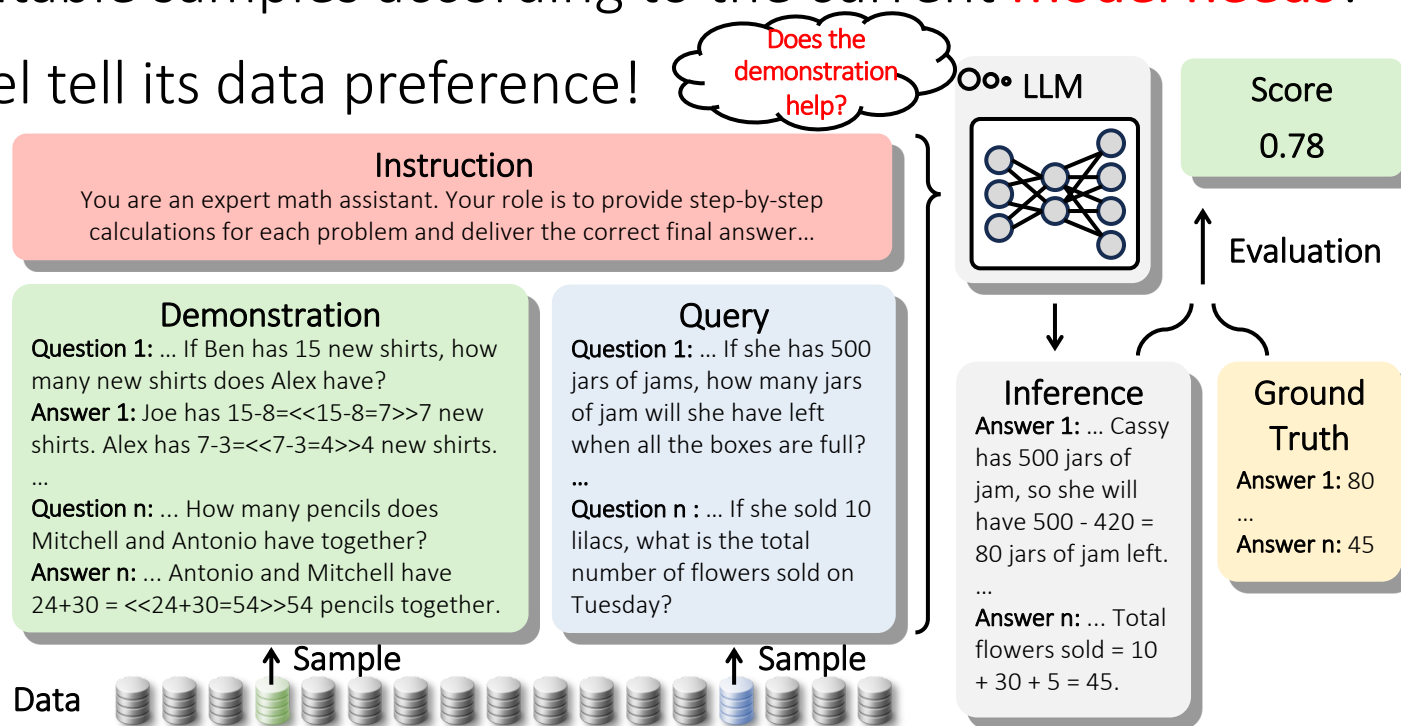


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- **Before fine-tuning**, let model tell its data preference!

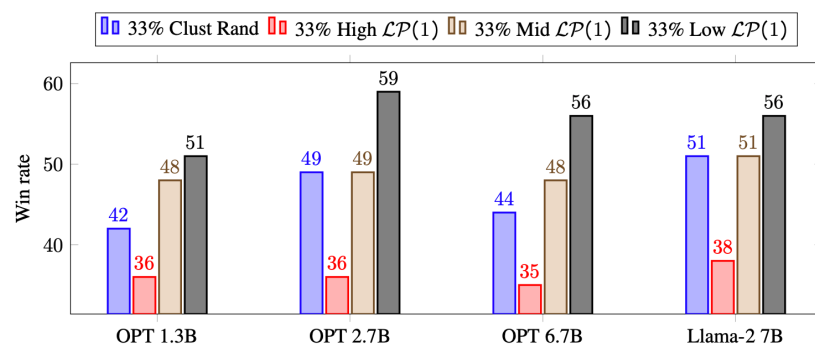
- In-context learning as a training-free selector:
Score samples through iteratively evaluating their contributions on queries [1]



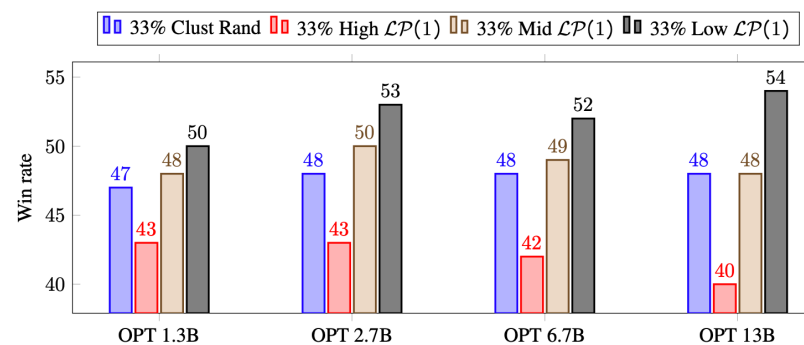
[1] Shaobo Wang, Xiangqi Jin, Ziming Wang, Jize Wang, Jiajun Zhang, Kaixin Li, Zichen Wen, Zhong Li, Conghui He, Xuming Hu, Linfeng Zhang: Data Whisperer: Efficient Data Selection for Task-Specific LLM Fine-Tuning via Few-Shot In-Context Learning. ACL (1) 2025: 23287-23305

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 - Optimization-related signals: Training dynamics, preference loss, and gradient-based valuation [1-3].



(a) Dolly dataset



(b) Alpaca-Data

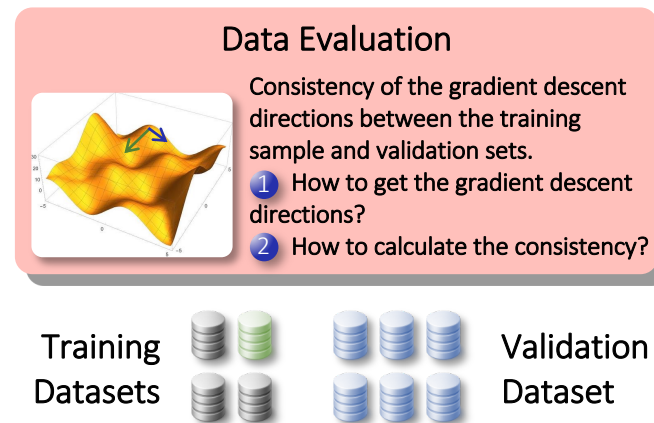
[1] Dheeraj Mekala, Alex Nguyen, Jingbo Shang: Smaller Language Models are capable of selecting Instruction-Tuning Training Data for Larger Language Models. ACL (Findings) 2024: 10456-10470

[2] Yang Wu, Huayi Zhang, Yizheng Jiao, Lin Ma, etc.: ROSE: A Reward-Oriented Data Selection Framework for LLM Task-Specific Instruction Tuning. EMNLP (Findings) 2025: 13200-13219

[3] Hyeonseok Moon, Jaehyung Seo, Seonmin Koo, Jinsung Kim, etc.: LimaCost: Data Valuation for Instruction Tuning of Large Language Models. EMNLP (Findings) 2025: 12841-12854

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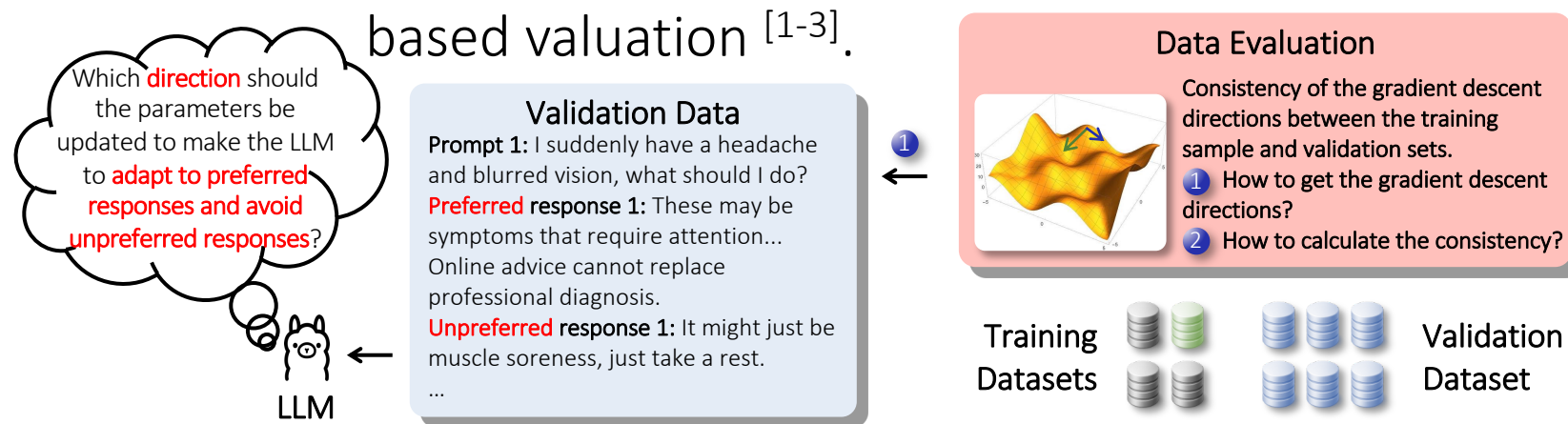
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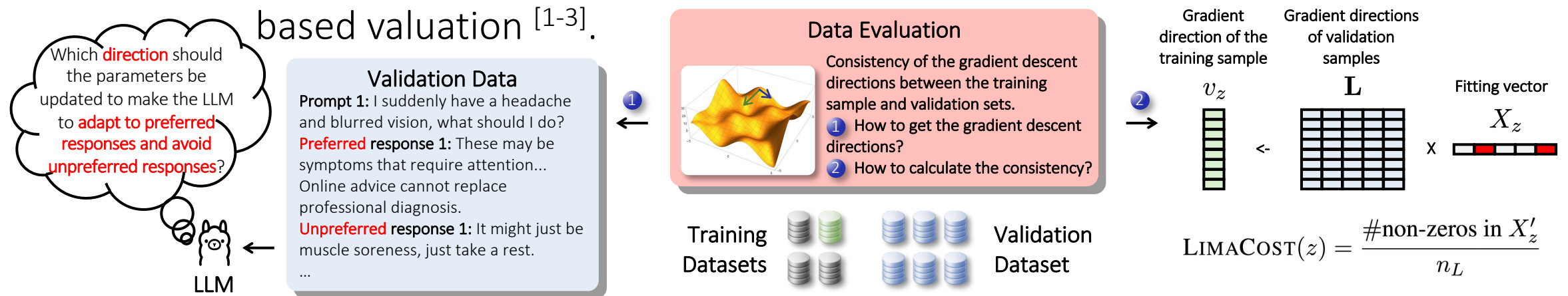
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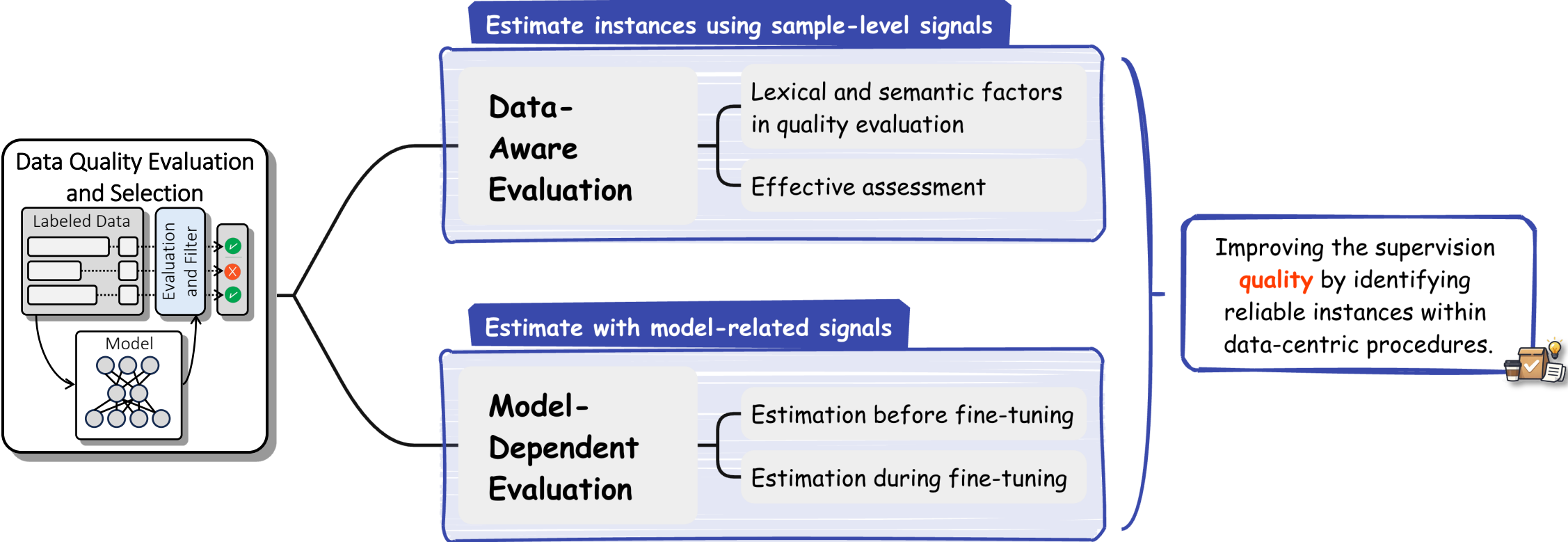


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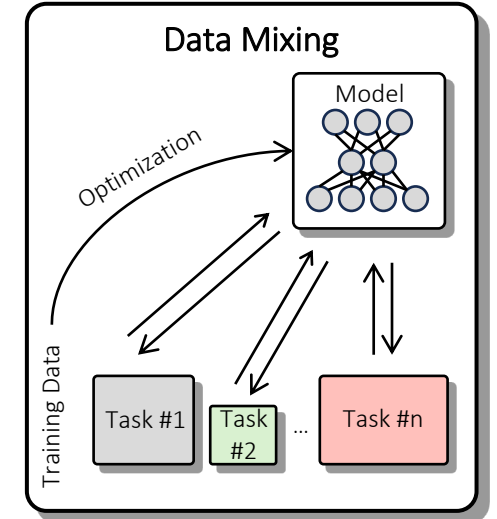
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Takeaways



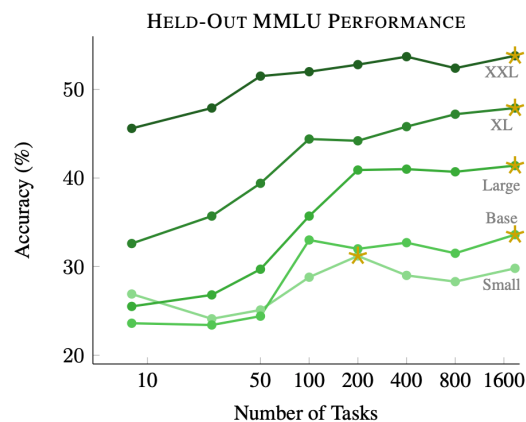
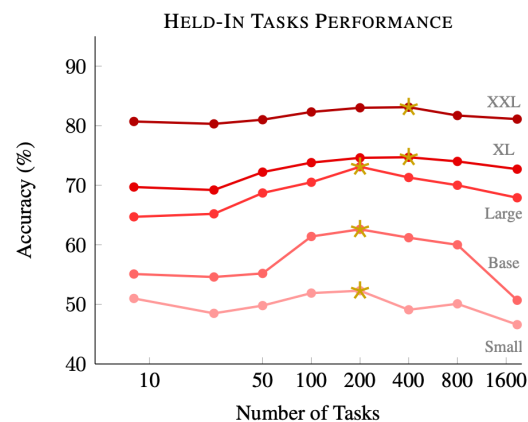
Data Mixing

- Data mixing addresses data scarcity from the perspective of data composition to enrich task-relevant supervision.
- Static mixing: Determine a fixed data composition before training
 - How to conduct a **mixture design** specifically for the target tasks?

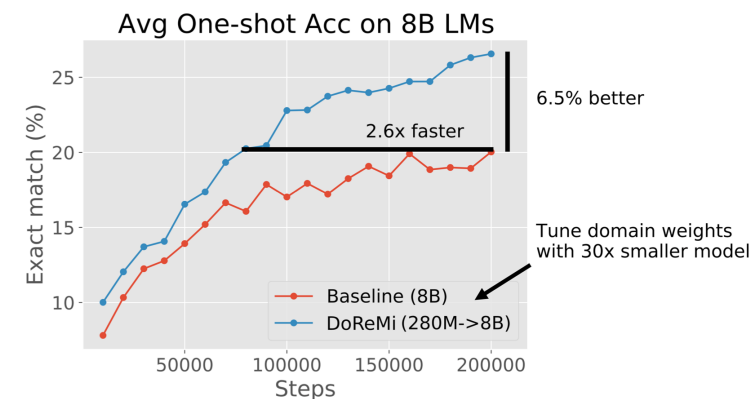


Data Mixing

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 - How to conduct a **mixture design** specifically for the target tasks?
 - Many earlier works have found that data composition can substantially affect convergence and final performance [1-2].



TRAIN MIXTURES	METRICS		
	Held-In	CoT	MMLU
All (Equal)	64.9	41.4	47.3
All - T0-SF	63.2	43.4	44.7
All - Flan 2021	55.3	38.6	45.7
All - Super-Nat. Inst.	65.9	42.2	46.8
All - CoT	65.6	29.1	46.8
All - Prog. Synth.	66.9	42.3	46.8
All - Dialog	65.4	40.3	47.1
All (Weighted)	66.4	40.1	48.1



[1] Shayne Longpre, Le Hou, Tu Vu, Albert Webson, Hyung Won Chung, Yi Tay, Denny Zhou, Quoc V. Le, Barret Zoph, Jason Wei, Adam Roberts: The Flan Collection: Designing Data and Methods for Effective Instruction Tuning. ICML 2023: 22631-22648

[2] Sang Michael Xie, Hieu Pham, Xuanyi Dong, Nan Du, Hanxiao Liu, Yifeng Lu, Percy Liang, Quoc V. Le, Tengyu Ma, Adams Wei Yu: DoReMi: Optimizing Data Mixtures Speeds Up Language Model Pretraining. NeurIPS 2023

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 - Irrelevant tasks may hurt target-task performance. **Careful task choices!**
 - Empirical study on low-resource MT: General data from other tasks is harmful for adaptation, but multi-lingual parallel data works ^[1]

SFT Mixture	chrF++
250K spa-X MT + 250K Aya (asm, mni)	26.91 ± 0.42
250K spa-X MT + 250K Alpaca (eng)	28.02 ± 0.41
500K spa-X MT only	29.57 ± 0.43

Method	All	HRL	LRL
Bilingual LLM-MT	14.51	24.55	10.75
Multilingual LLM-MT	18.73	23.59	16.90

[1] Vivek Iyer, Bhavitvya Malik, Pavel Stepachev, Pinzhen Chen, Barry Haddow, Alexandra Birch: Quality or Quantity? On Data Scale and Diversity in Adapting Large Language Models for Low-Resource Translation. WMT 2024: 1393-1409

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 - Task selection for task-specific adaptation: Select tasks that contributes most without training samples support [2]

“ This work leverages instruction similarity scores to determine task relevance and select informative tasks for training...We discover that this simplified instruction-only selection method adequately identifies related tasks and significantly improves performance. ”

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[2] Changho Lee, Janghoon Han, Seonghyeon Ye, Stanley Jungkyu Choi, Honglak Lee, Kyunghoon Bae: Instruction Matters: A Simple yet Effective Task Selection for Optimized Instruction Tuning of Specific Tasks. EMNLP 2024: 18620-18642

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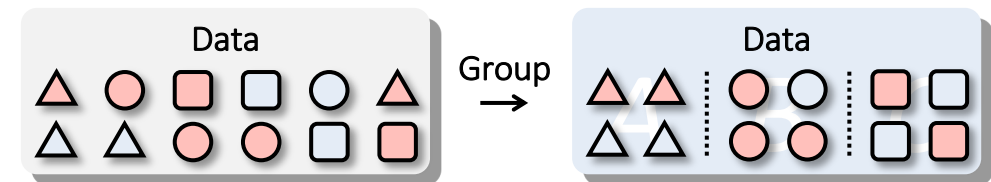
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 - Inside the chosen tasks, the **sample selection and organization** also matter!
 - Instance selection as an optimization problem: Use submodular optimization to select tasks and non-redundant samples within each task ^[1]

	Submodular Function	$f(X)$
Models Representation	→ Facility Location	$\sum_{i \in \mathcal{V}} \max_{j \in X} s_{ij}$
Models a Trade-Off Between Representation and Diversity	→ Graph Cut	$\sum_{i \in \mathcal{V}, j \in X} s_{ij} - \lambda \sum_{i, j \in X} s_{ij}$
Models Diversity	→ Log Determinant	$\log \det(\mathcal{S}_X)$

[1] H. S. V. N. S. Kowndinya Renduchintala, Sumit Bhatia, Ganesh Ramakrishnan: SMART: Submodular Data Mixture Strategy for Instruction Tuning. ACL (Findings) 2024: 12916-12934

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Group similar data in the same training batch to help LLMs learn commonalities more clearly ^[2]

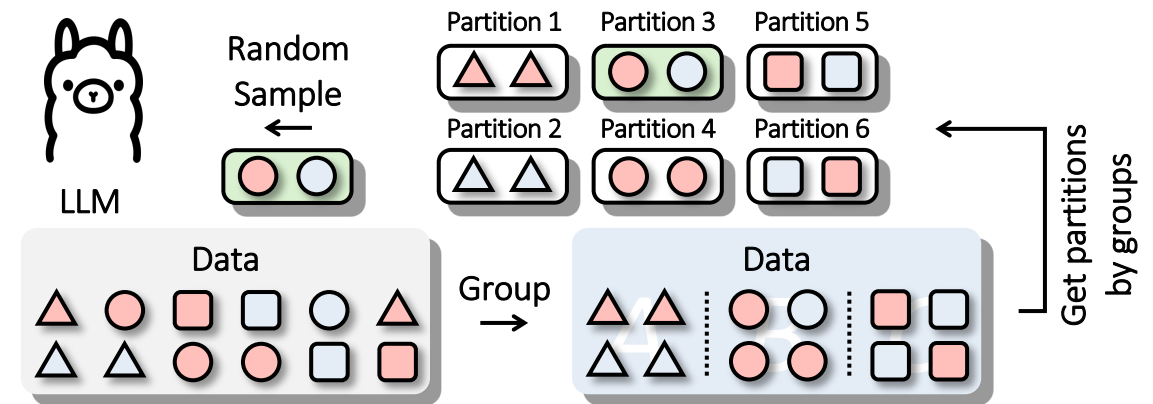


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[2] Jun Rao, Xuebo Liu, Lian Lian, Shengjun Cheng, Yunjie Liao, etc.: CommonIT: Commonality-Aware Instruction Tuning for Large Language Models via Data Partitions. EMNLP 2024: 10064-10083

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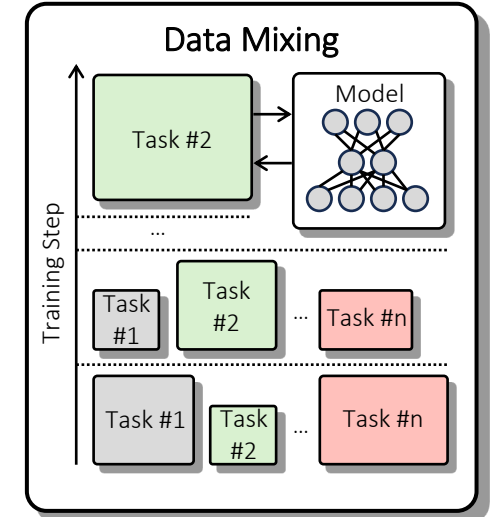


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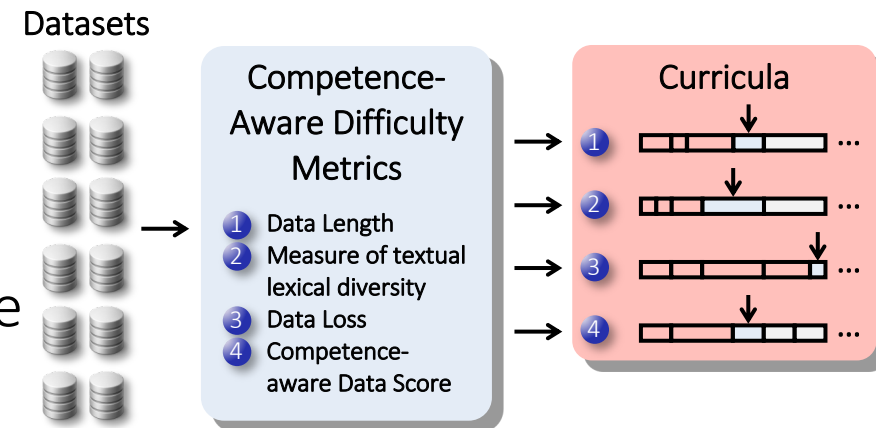
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- Curriculum-learning-based methods: Dynamically organizing sub-curricula and training schedules with multiple views and competence-aware scheduling ^[1]

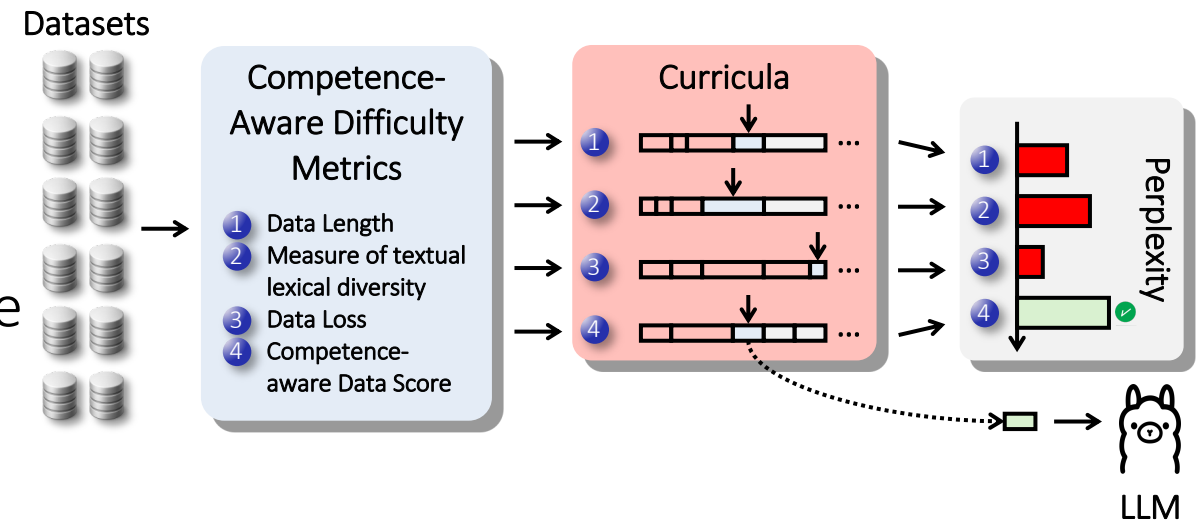


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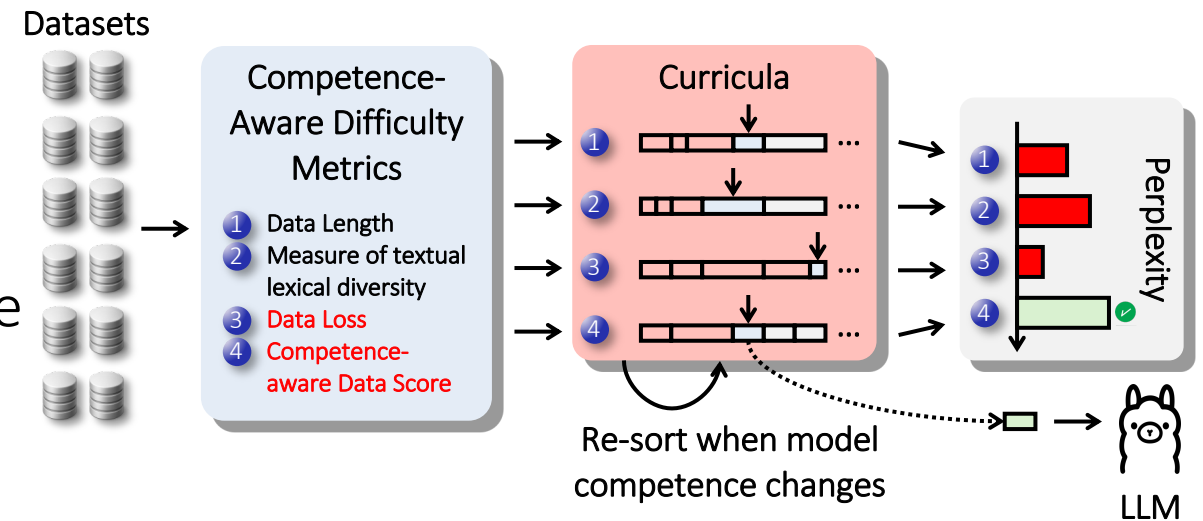


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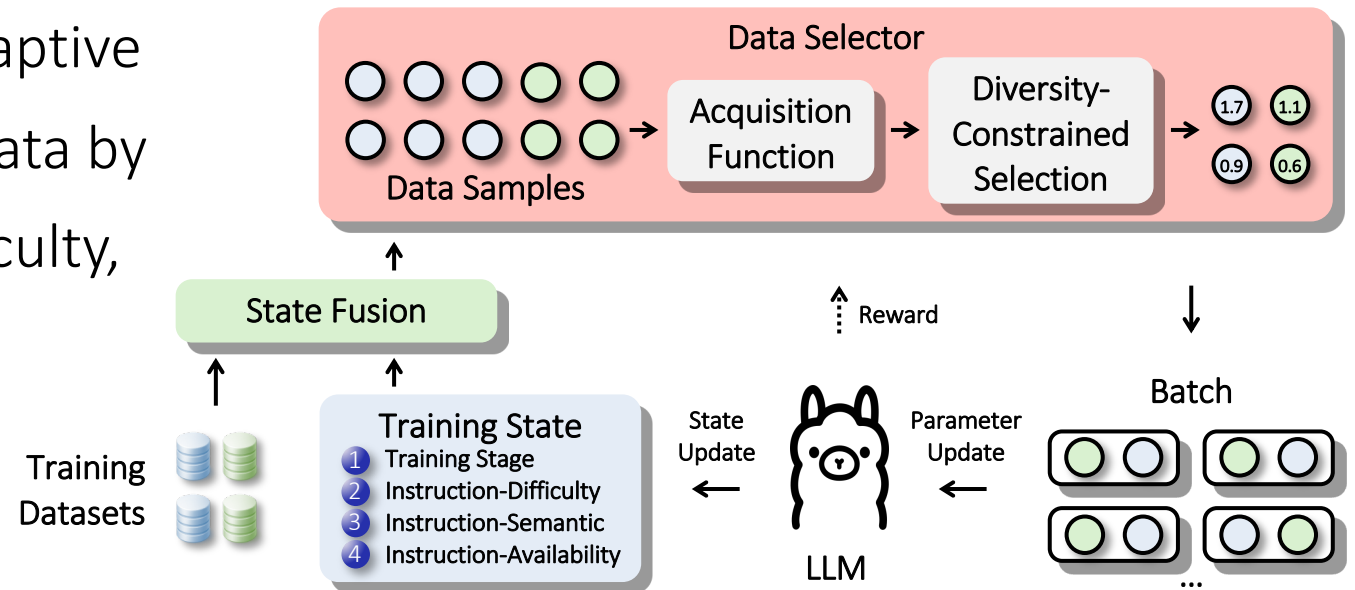
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 - RL-based methods: Adaptive reweight and sample data by the transferability, difficulty, availability, etc. [1-2]

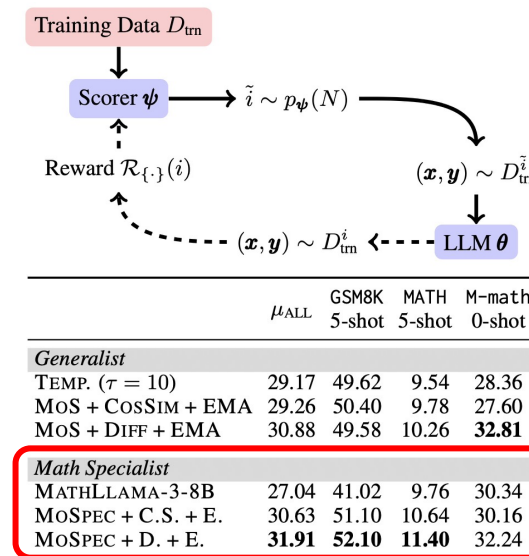


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[2] Minghao Wu, Thuy-Trang Vu, Lizhen Qu, Reza Haf: Mixture-of-Skills: Learning to Optimize Data Usage for Fine-Tuning Large Language Models. EMNLP 2024: 14226-14240

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 - Also support adaptation to particular target tasks [2]

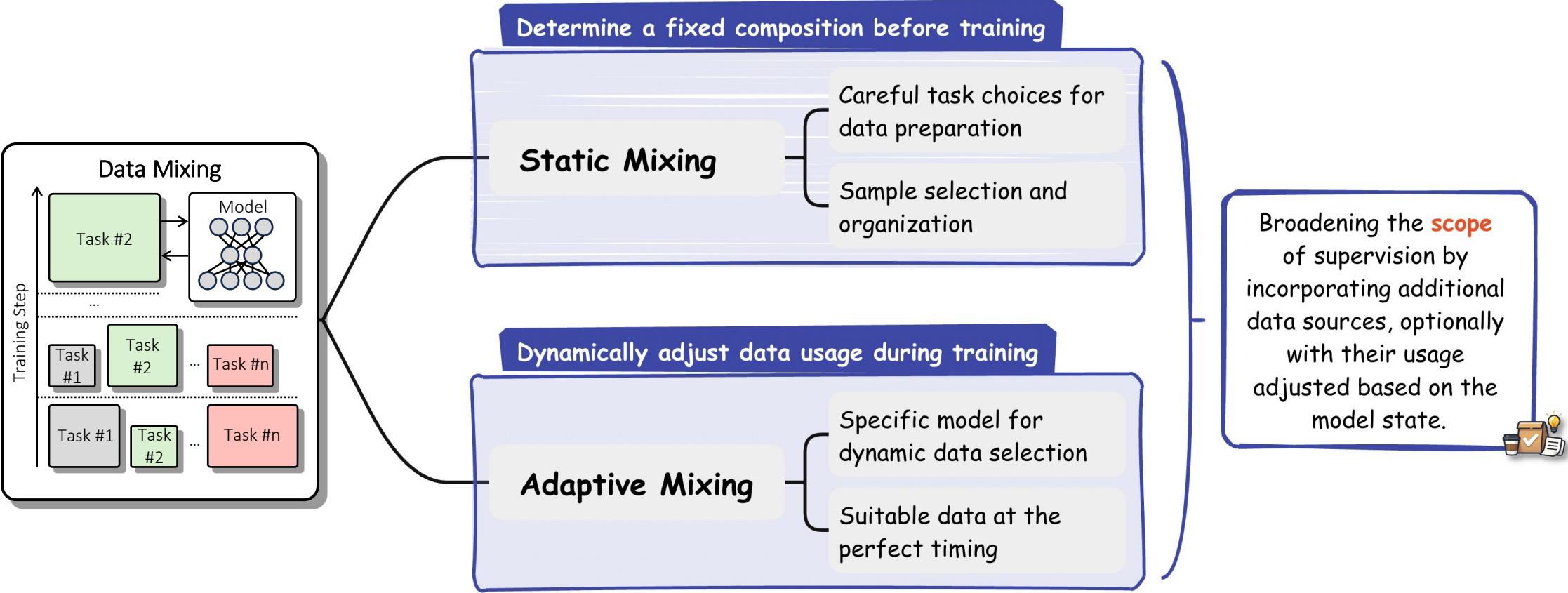


“ For MOSPEC with COSSIM, we compute the cosine similarity between *Mathematics* and other datasets, including *Mathematics* itself. For MOSPEC with DIFF, we double the reward for the *Mathematics* dataset. ”

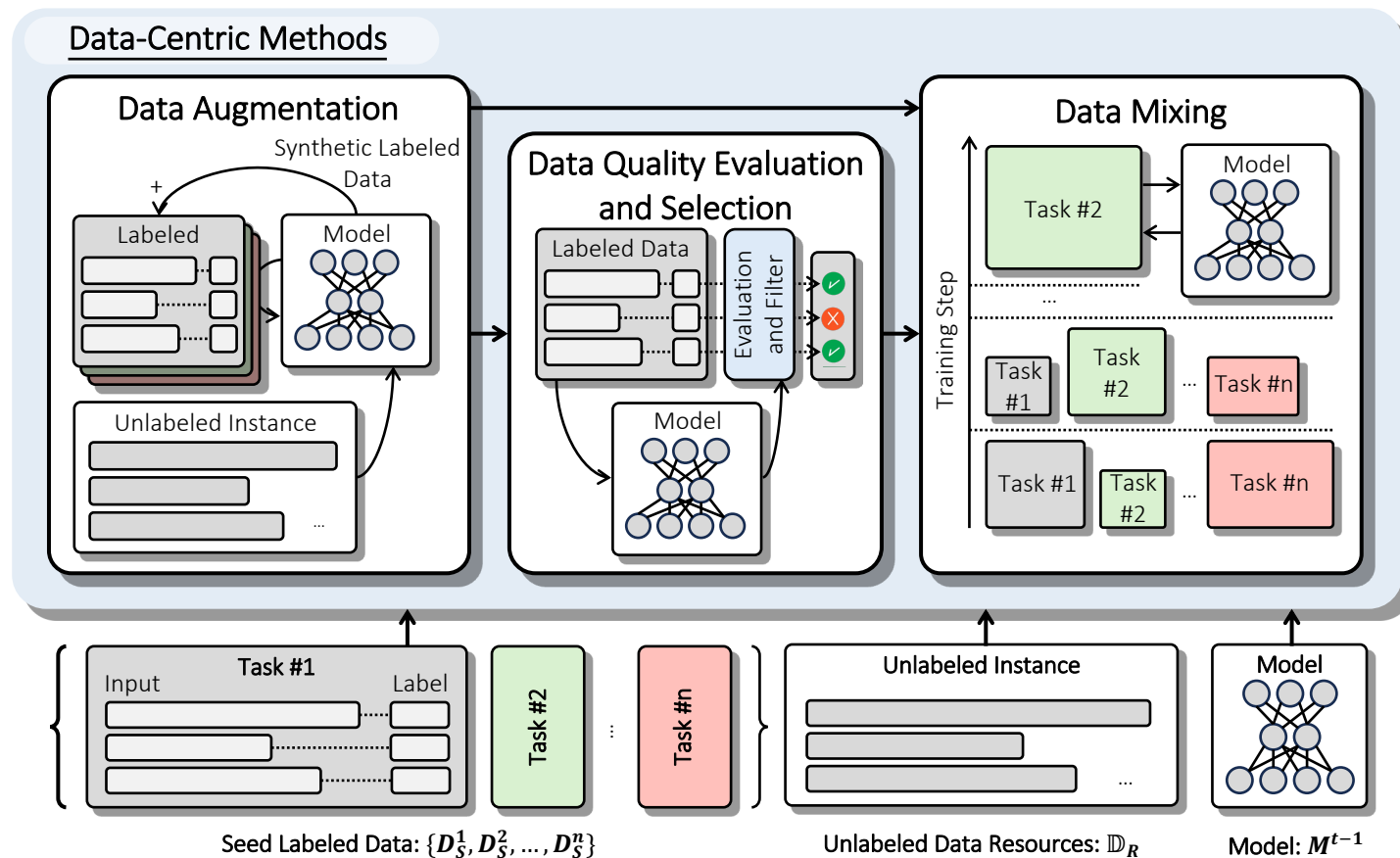
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Takeaways



Summary



Wrap up of this lecture!

- Understand the main angles to address data scarcity from a data-centric perspective ★☆☆
- Learn how data augmentation, data quality evaluation and selection, and data mixing methods can support effective low-resource task adaptation ★★★



Summary

Work	Type	Method Input			Experimental Setting		
		Data Source	Auxiliary Model	Unlabeled Data	Task	Dataset	Foundation Model
STAR (Ma et al., 2024)	Data Augmentation	Single	✓	×	Event Extraction; Relation Extraction	ACE05; TACRED	GPT-3.5; GPT-4
CoDa (Evuru et al., 2024)	Data Augmentation	Single	✓	×	Text Classification; NER; QA	HuffPost; Yahoo; OTS; ATIS; Massive; CoNLL-2003; OntoNotes-5.0; EBMNLP; BC2GM; SQuAD; NewsQA	LLaMA-13B; Mistral-7B; LLaMA-7B
Prompting-Based Synthetic Data Generation (Schmidt et al., 2024b)	Data Augmentation	Single	✓	×	QA	Few-Shot MRQA Benchmark	T5 v1.1-large
DawGen (Guo et al., 2024)	Data Augmentation	Single	✓	×	Text Classification; Semantic Understanding	QQP; MRPC; MNLI; SNLI; QNLI; RTE; SICK	CTRL-1.6B; T5-large; Flan-T5-large
SEMIEVOL (Luo et al., 2025a)	Data Augmentation	Single	✓	✓	QA; Reasoning; Sentiment Analysis	MMLU; MMLU-Pro; ARC; FPB; USMLE; PubMedQA; ConvFinQA	GPT-4o-mini; LLaMA-3-8B
VerifyMatch (Park and Caragea, 2024)	Data Augmentation	Single	✓	✓	Text Classification	RTE; SICK; SNLI-2.5k; MNLI-2.5k	LLaMA-3-8B-Instruct; BERT-base
SNARE (Parekh et al., 2025)	Data Quality Evaluation	Single	×	✓	Event Detection	ACE; SPEED; GENIA	LLaMA-3-8B-Instruct; LLaMA-3-70B-Instruct; GPT-3.5
Data Whisperer (Wang et al., 2025d)	Data Quality Evaluation	Single	✓	×	QA; Summarization; Reasoning	BioInstruct; DialogSum; GSM8K	LLaMA-3-8B-Instruct; Qwen-2.5-7B-Instruct; Mistral-Nemo-Instruct-2407; Qwen-2.5-3B-Instruct; Mistral-7B-Instruct-v0.2
ROSE (Wu et al., 2025)	Data Quality Evaluation	Single	✓	×	Preference Alignment	SHP; Stack Exchange; HH-RLHF	LLaMA-2-7B; LLaMA-2-13B; LLaMA-3.1-8B; LLaMA-3.1-8B-Instruct
CLMA (Liu et al., 2025c)	Data Quality Evaluation	Single	✓	✓	Relation Extraction	SemEval-2010; TACRED; TACREV; Re-RACRED	GPT-3.5; Flan-T5-large
INSTA (Lee et al., 2024)	Data Mixing	Multiple	×	×	General Benchmarks; Text Classification	P3; NIV2; BIG-Bench; BIG-Bench Hard	T5-3B
MoSPEC (Wu et al., 2024)	Data Mixing	Multiple	✓	×	QA; Reasoning; Dialogue	MMLU; MT-Bench; GSM8K; MATH; M-Math	Qwen-1.5-0.5B; Gemma-2B; LLaMA-3-8B

Wrap up of this lecture!

- Understand the main angles to address data scarcity from a data-centric perspective ★☆☆
- Learn how data augmentation, data quality evaluation and selection, and data mixing methods can support effective low-resource task adaptation ★★★



Exercises

Task 1

Do data augmentation, data quality evaluation and selection, and data mixing cover all major directions of data-centric methods? Are there any other areas that also contribute to the low-resource issue?

Task 2

Some methods combine multiple strategies to mitigate scarce supervision, and the boundaries between categories are not always mutually exclusive. Please give an example and share your thoughts.

Task 3

Many data-centric methods are open-sourced. Can you implement one of them and try to reproduce its experimental results?

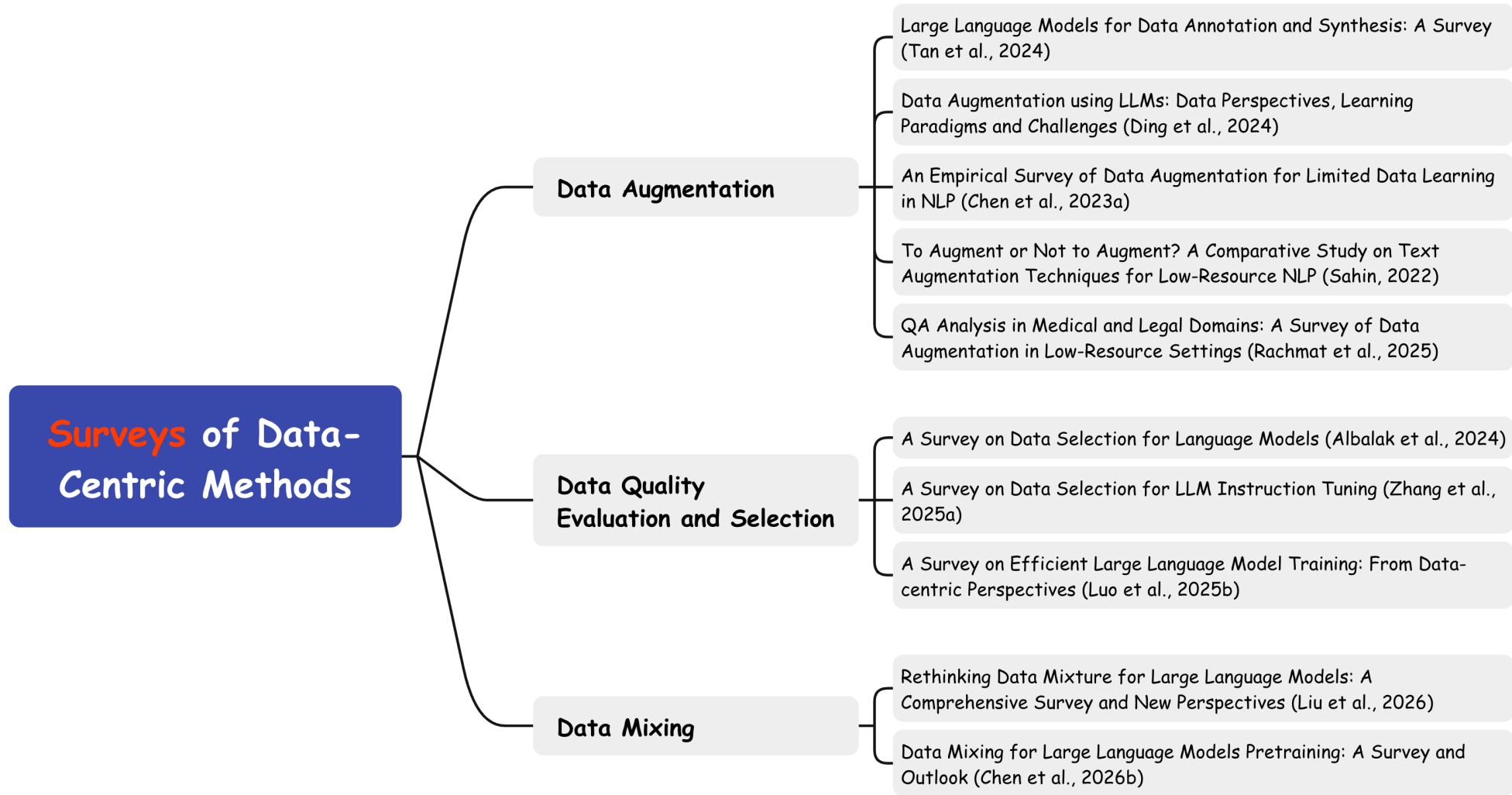


Please feel free to submit your exercise results and feedback to our GitHub repository (<https://github.com/liyingqiao2012/Low-Resource-Task-Adaptation-in-LLMs/issues>).

exercise

discussion

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Irish Gaelic

Go raibh
maith agat!

Cantonese

多謝!

Breton

Trugarez
deoc'h!

Manchu

ᡩᠠᡳᠨᡤᡠᡵᡠ!

Large Language Model

Adyghe

Тхьауе-
гъэпсэу!

LLM with Strong Low-Resource Languages Support



>>

“Article 2 – All individuals are entitled to the rights
declared without discrimination based on language.”
- Universal Declaration of Human Rights (1948)

United Nations
(Eleanor Roosevelt)

Presenter
Video